Kai Wu

Postdoctoral Researcher in Computational Astrophysics

Astronomisches Rechen-Institut, Zentrum für Astronomie, Heidelberg University, Germany

kai.wu@uni-heidelberg.de | +49 (0)176 6592 9094 | GitHub | ORCID | Google Scholar | NASA ADS | Personal Site

RESEARCH PROFILE

Computational astrophysicist specializing in collisional stellar dynamics and direct N-body simulations of dense stellar systems. Research spans massive star clusters, compact-object populations, Population III clusters, and the dynamical evolution of planetary systems in cluster environments. Developer and code administrator of NBODY6++GPU, with experience in million-body simulations, heterogeneous GPU supercomputing, and large-scale simulation-data analysis.

ACADEMIC APPOINTMENT

Postdoctoral Researcher, Astronomisches Rechen-Institut (ARI-ZAH), Heidelberg University, Germany **2024-present**

Research group of apl. Prof. Rainer Spurzem

- Lead and analyse direct N-body simulations of globular, nuclear, and Population III star clusters within the DRAGON-III programme.
- Develop coupled methods for planetary-system evolution in dense stellar environments using NBODY6++GPU and REBOUND.
- Maintain scientific software and simulation workflows across heterogeneous CPU/GPU supercomputing systems.

EDUCATION

PhD in Mathematical Sciences, University of Liverpool, UK **2019-2024**

Thesis: Dynamical stability of debris discs with planets in star clusters

- Supervisor: Prof. M. B. N. Kouwenhoven; research conducted off-site at Xi'an Jiaotong-Liverpool University (XJTLU).
- Full scholarship covering tuition and stipend.

BSc in Physics, Nankai University, China **2015-2019**

Thesis: Dynamical evolution of young open clusters in the solar neighbourhood

- Supervisor: Prof. Fang Bo; cumulative GPA: 85.3/100.
- Graduation thesis awarded Excellent Thesis (top 3%).

RESEARCH AND SOFTWARE CONTRIBUTIONS

DRAGON-III and dense stellar systems: million-body simulations of globular and nuclear star clusters over cosmic time; rotating and extremely massive Population III clusters; compact objects and multi-messenger observables.

Planetary systems in clusters: dynamical stability of debris discs and planets, survival of planetary architectures along stellar orbits, and long-term evolution of free-floating minor bodies.

NBODY6++GPU development and administration: code administrator since 2022; introduced Git-based development and CI testing; manage pull requests, issues, documentation, and user support.

Numerical-method development: implemented Fortran routines for planetary-system evolution and C/C++ interfaces for high-performance management of REBOUND simulations.

HPC and data: simulation and benchmarking experience on JUWELS, LUMI, Dongfang, Silkroad, and Kepler; CUDA/HIP platforms; analysis of more than 30 TB of DRAGON simulation data.

PUBLICATIONS

Peer-reviewed journal articles

[1] **Kai Wu**, Ataru Tanikawa, Francesco Flammini Dotti, Marcelo C. Vergara, Vahid Amiri, Boyuan Liu, Albrecht W. H. Kamlah, Manuel Arca Sedda, Nadine Neumayer, and Rainer Spurzem. "Direct N-body simulations of rotating and extremely massive Population III star clusters." *Astronomy & Astrophysics*, 709, A142 (2026). doi:10.1051/0004-6361/202558659

[2] Francesco Flammini Dotti, M. B. N. Kouwenhoven, **Kai Wu**, Abbas Askar, Peter Berczik, Mirek Giersz, Rainer Spurzem, and Ian Dobbs-Dixon. "Dynamical evolution of massless particles in star clusters with NBODY6++GPU-MASSLESS: II. The long-term evolution of free-floating comets." *Astronomy & Astrophysics*, 706, A219 (2026). doi:10.1051/0004-6361/202557334

[3] Guimei Liu, Yu Zhang, Jing Zhong, Li Chen, Xiangcun Meng, and **Kai Wu**. "Formation and evolution of new primordial open cluster groups: Feedback-driven star formation." *Astronomy & Astrophysics*, 696, A117 (2025). doi:10.1051/0004-6361/202452774

[4] Songmei Qin, Jing Zhong, Tong Tang, Yueyue Jiang, Long Wang, **Kai Wu**, Friedrich Anders, Lola Balaguer-Nunez, Guimei Liu, Chunyan Li, Jinliang Hou, and Li Chen. "Unveiling the binary nature of NGC 2323." *Astronomy & Astrophysics*, 693, A317 (2025). doi:10.1051/0004-6361/202452962

[5] **Kai Wu**, M. B. N. Kouwenhoven, Rainer Spurzem, and Francesco Flammini Dotti. "Influence of planets on debris discs in star clusters - II. The impact of stellar densities." *Monthly Notices of the Royal Astronomical Society*, 533(4) (2024). doi:10.1093/mnras/stae2067

[6] **Kai Wu**, M. B. N. Kouwenhoven, Rainer Spurzem, and Xiaoying Pang. "Influence of planets on debris discs in star clusters - I. The 50 au Jupiter." *Monthly Notices of the Royal Astronomical Society*, 523(4) (2023). doi:10.1093/mnras/stad1673

[7] M. B. N. Kouwenhoven et al. (including **Kai Wu**). "Planetary systems in dense stellar environments." *Journal of Physics: Conference Series*, 1523, 012011 (2020). doi:10.1088/1742-6596/1523/1/012011

Accepted conference proceeding

[8] **Kai Wu**, Philip Cho, Rainer Spurzem, Long Wang, Francesco Flammini Dotti, and Vahid Amiri. "DRAGON-III simulations: modelling million-body globular and nuclear star clusters over cosmic time." *Proceedings of IAU Symposium 398*, accepted

Manuscripts in preparation

[9] **Kai Wu**, Francesco Flammini Dotti, Marcelo C. Vergara, Vahid Amiri, and Rainer Spurzem. "DRAGON-III simulation - I. Multi-messenger observations in the computer and comparison with observed astrophysical objects." . *In preparation*

[10] **Kai Wu**, M. B. N. Kouwenhoven, Rainer Spurzem, and Francesco Flammini Dotti. "Linking the survivability of planets in star clusters with the orbits of their host stars." . *In preparation*

HONOURS, AWARDS, AND COMPETITIVE SUPPORT

Excellent Teaching Assistant Award (top 2%), XJTLU	**2023**
Best Oral Presentation Award (top 5%), XJTLU Postgraduate Research Symposium	**2022**
Best Poster Award (top 5%), XJTLU Postgraduate Research Symposium	**2021**
Full-scholarship PhD, tuition waiver and stipend, XJTLU	**2019-2023**
Excellent Graduation Thesis (top 3%), Nankai University	**2019**
Selected Boling and university scholarships, Nankai University	**2015-2017**
Second Prize (top 10%), Nankai Physicists' Tournament	**2016**

INVITED LECTURES AND SEMINARS

ARI-ZAH, Heidelberg University: From Debris Disks to Million-Body Clusters: Planetary Perturbations, Stellar Interactions, and DRAGON-III	**Jul. 2025**
Zhejiang University: Dynamics of debris discs with planets in star clusters	
KIAA-Peking University: Dynamics of debris discs in star clusters	**Jul. 2023**
Shanghai Astronomical Observatory: An in-depth introduction to numerical star-cluster simulations,	**Jul. 2023**
Shanghai Astronomical Observatory: 30-hour doctoral course of N-body simulations with NBODY6++GPU	**Aug. 2023**
Spurzem's International seminar: Git and GitHub for NBODY6++GPU development and collaboration	**Apr. 2022**

SELECTED CONFERENCE PRESENTATIONS

Annual Meeting of the German Astronomical Society, Goerlitz - two contributed talks on DRAGON-III and planetary systems in star clusters	**Sep. 2025**
European Astronomical Society Annual Meeting, Cork - Dynamical stability of debris discs with planets in star clusters	**Jun. 2025**
IAU Symposium 398 / MODEST-25, Seoul - DRAGON-III million-body globular and nuclear star clusters	**Jun. 2025**
Gas Accretion in Planet Formation, MPIA - Star clusters, protoplanetary discs, and debris-disc evolution	**Mar. 2025**
12th Kazakhstan-China-Korea Workshop, Astana - Dynamical stability of debris discs with planets in star clusters	**May 2024**
SPP 1992 Final Colloquium, Berlin - Dynamical evolution of planetary systems in star clusters	**Mar. 2024**
Star Cluster Workshop: From Observations to Simulations, Suzhou - Debris discs and open-cluster evolution	**Oct. 2023**
Annual Meeting of the Chinese Astronomical Society, Weihai - Cluster environments and debris discs	**Aug. 2023**
Structure, Formation and Evolution of Star Clusters, Sun Yat-sen University - Planetary systems in clusters	**Jun. 2023**
5th Young Scientist Forum of Planetary Science, Sanya - The 50 au Jupiter	**Mar. 2023**
PFE-SPP 1992 Joint Meeting, online - Planets and debris particles in star clusters	**Sep. 2023**

TEACHING AND ACADEMIC TRAINING

Associate Fellow of the Higher Education Academy (AFHEA), Advance HE — **2024**
- Teaching qualification aligned with the UK Professional Standards Framework.

Part-time Lecturer - IB Calculus, Suzhou Research Institute of Xi'an Jiaotong University — **Jan.-Jul. 2022**
- Designed and delivered 105 hours of calculus instruction for an international high-school preparatory programme.

Teaching Assistant / Tutor, Xi'an Jiaotong-Liverpool University — **2019-2023**
- Delivered approximately 30 hours of English-language teaching in physics experiments, calculus, advanced linear algebra, and statistical methods.
- Received the School of Mathematics and Physics Excellent Teaching Assistant Award (top 2%).

Invited Instructor - Doctoral N-body Course, Shanghai Astronomical Observatory — **Aug. 2023**
- Delivered a 30-hour practical course on simulating star clusters with NBODY6++GPU.

Invited Tutorial Instructor, International NBODY6++GPU seminar — **Apr. 2022**
- Delivered a four-hour tutorial on Git, GitHub, and collaborative scientific-software development.

ACADEMIC SERVICE AND MEMBERSHIPS

Referee, Astronomy & Astrophysics	**2026-present**
Referee, Planetary and Space Science	**2025-present**
Member, COST Action CA22133 "The Birth of Solar Systems"	**2023-present**
Member, Silk Road Project for stellar dynamics and computational astrophysics	**2019-present**

SELECTED ACADEMIC VISITS

National Astronomical Observatories of China and KIAA-Peking University - visitor of Dr. Shuo Li	**Dec. 2024; Jun. 2025;**

Frankfurt Institute for Advanced Studies - visitor of Prof. Volker Lindenstruth **Jan. 2025**
Heidelberg University - guest researcher funded by DFG SPP 1992 **Mar.-Apr. 2024**
National Astronomical Observatories of China - visiting scholar funded by XJTLU **Dec. 2019**
CITA, University of Toronto - visiting student of Prof. Yanqin Wu **Jul.-Oct. 2018**
Space Telescope Science Institute / Johns Hopkins University - visiting student of Prof. Christine Chen **Jun.-Sep. 2017**
Earlier visits: Purple Mountain Observatory; National University of Singapore; Nanyang Technological University **2016**

TECHNICAL EXPERTISE

Scientific programming: Python, Fortran, C/C++, Bash; NumPy, SciPy, pandas, multiprocessing
Parallel and GPU computing: MPI, OpenMP, CUDA, HIP; heterogeneous CPU/GPU systems
Scientific software: NBODY6++GPU, PeTar, REBOUND, MERCURY
Data and visualisation: HDF5, Matplotlib, Plotly; large-scale simulation-data processing
Development workflows: Git, GitHub, CI/CD, Docker, documentation, issue and release management
Typesetting and reproducibility: LaTeX, Linux-based research workflows

ACADEMIC REFERENCES

Prof. Mattheus Bartholomeus Nicolaas Kouwenhoven t.kouwenhoven@xjtlu.edu.cn
Professor; Head of department at XJTLU
Mathematical Sciences Department, Xi’an Jiaotong- Liverpool University, Xi'an Jiaotong-Liverpool University (XJTLU)
Prof Rainer Spurzem spurzem@ari.uni-heidelberg.de spurzem@nao.cas.cn
apl. Professor at Astronomisches Rechen-Institut: Zentrum für Astronomie (ARI-ZAH) in University of Heidelberg
Research Professor at National Astronomical observatory of China (NAOC)
Professor at Kavli Institute for Astronomy and Astrophysics at Peking University (KIAA-PKU)
Prof. Peter Berczik berczik@camk.edu.pl
Professor at Nicolaus Copernicus Astronomical Centre of the Polish Academy of Sciences,
Dr. Francesco Flammini Dotti ff2415@nyu.edu
Fellow post-doctoral researcher at the Department of Physics and the Center of Astrophysics and Space science, New York University Abu Dhabi